

ANAL KUMAR BISWAS

Senior Software Engineer | Distributed Systems | Cloud Infrastructure | Backend Engineering

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6 years of experience designing, developing, and operating cloud-native distributed systems, event-driven microservices, backend platforms, and enterprise applications across AWS and Google Cloud Platform (GCP). Strong experience with Kubernetes, Terraform, Docker, Kafka, Python, Node.js, Java, PostgreSQL, MongoDB, Redis, and observability platforms. Experienced in system design, scalable architecture, asynchronous processing, performance optimization, production reliability, identity and access management, cloud infrastructure, CI/CD, and developer automation. Delivered systems supporting 10,000+ concurrent users, 1M+ daily events, 100,000+ managed assets, and 5,000+ enterprise users.

CORE TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript

Backend Development: Python FastAPI, Node.js, NestJS, Express.js, Java

Distributed Systems: Microservices, Event-Driven Architecture, Kafka, Asynchronous Processing, Idempotency, Retry Mechanisms, Dead Letter Queues, System Design

Cloud Platforms: AWS (EKS, EC2, ECS, S3, Lambda etc.), Google Cloud Platform (GCP)

Containers and Infrastructure: Kubernetes, Docker, Terraform, Infrastructure as Code

Databases: PostgreSQL, MongoDB, Redis, OpenSearch

Observability and Reliability: OpenTelemetry, Distributed Tracing, Metrics, Logging, Horizontal Pod Autoscaling (HPA), Production Troubleshooting

Security and Identity: Keycloak, OAuth 2.0, OpenID Connect, RBAC, Microsoft Entra ID, Single Sign-On

AI and Developer Automation: Model Context Protocol (MCP), Claude Code, LangChain

Frontend Technologies: React, Next.js, Angular, Electron

CI/CD and DevOps: Jenkins, CI/CD Pipelines, Terraform, Kubernetes, Cloud Deployment

PROFESSIONAL EXPERIENCE

GNAPI TECHNOLOGIES PVT. LTD.

Senior Software Engineer | July 2025 - Present

Project: AI-Based Live Broadcasting Platform

Project Overview: High-throughput live broadcasting and media platform designed to deliver real-time video streams to large numbers of concurrent viewers. The platform was migrated from legacy streaming processes to a WebRTC and LiveKit-based architecture running on AWS Kubernetes infrastructure.

Key Contributions:

- Owned the system design and architecture migration from legacy streaming workflows to WebRTC and LiveKit on Amazon EKS.
- Designed Kubernetes-based deployment architecture with Horizontal Pod Autoscaling to support variable traffic during live events.
- Worked across AWS cloud infrastructure for application deployment, compute, storage, networking, and supporting services.
- Scaled the platform to support 10,000+ concurrent viewers and 50+ simultaneous live camera feeds.
- Maintained 99.99% platform availability during peak tournament traffic.
- Worked on production scalability, infrastructure reliability, deployment architecture, and performance optimization.

Technologies: AWS, Amazon EKS, Kubernetes, Docker, WebRTC, LiveKit, HPA

Project: Enterprise QA Automation and Event-Driven Platform

Project Overview: Enterprise QA automation platform consisting of multiple backend services responsible for asynchronous test execution, event processing, telemetry collection, and automated test workflows.

Key Contributions:

- Led the transition from synchronous service workflows to an event-driven microservices architecture consisting of 15+ services.
- Designed Kafka-based asynchronous processing workflows using NestJS and TypeScript.
- Designed event schemas and service-to-service communication contracts.
- Implemented idempotency mechanisms to prevent duplicate event processing.
- Implemented retry strategies and Dead Letter Queues for failed message processing.
- Built centralized observability using OpenTelemetry for distributed tracing, metrics, and production diagnostics.
- Processed 1M+ telemetry events per day.
- Reduced operational latency by 70% and improved Mean Time to Recovery (MTTR) by 40%.

Technologies: Node.js, NestJS, TypeScript, Kafka, Microservices, OpenTelemetry, Kubernetes, AWS

Project: Spatial Asset Management System

Project Overview: Geospatial infrastructure management platform used to track and manage electrical infrastructure assets and their lifecycle through interactive mapping and spatial data services.

Key Contributions:

- Designed backend services using Python FastAPI, GeoServer, and MongoDB.
- Worked with GIS teams to design APIs and spatial data workflows.
- Provisioned and managed cloud infrastructure using Terraform.
- Supported real-time lifecycle tracking for 100,000+ electrical infrastructure assets.
- Optimized geospatial database queries and API performance.
- Improved map rendering performance by approximately 50%.

Technologies: Python, FastAPI, MongoDB, GeoServer, Terraform, AWS, GIS

Project: AI Developer Automation Agents

Project Overview: Internal developer productivity platform using AI-powered automation to provision development environments, manage dependencies, diagnose build failures, and automate repetitive engineering workflows.

Key Contributions:

- Engineered developer automation workflows using Claude Code and Model Context Protocol (MCP).
- Integrated AI agents with orchestrated Ubuntu-based development environments.
- Automated environment provisioning and dependency management.
- Automated build failure diagnostics and developer troubleshooting workflows.
- Reduced local development environment setup time from approximately 4 hours to less than 15 minutes.

Technologies: Python, MCP, Claude Code, Ubuntu, AI Agents, Automation

XOTIV TECHNOLOGIES PVT. LTD.

Full Stack Developer | December 2023 - June 2025

Project: DEVEBOL - AI Sports Education Platform

Project Overview: AI-powered sports education and analytics platform designed to support interactive sports learning and real-time usage during live sporting events.

Key Contributions:

- Designed backend services using Python FastAPI and Next.js.
- Integrated LangChain-based AI workflows into application services.
- Implemented Redis for caching and session state management.
- Reduced database load through caching and optimized application-level data access.
- Supported 500+ requests per second during live sports events.
- Reduced API response time by approximately 300 milliseconds.

Technologies: Python, FastAPI, Next.js, LangChain, Redis, PostgreSQL

Project: ReadMyRhythm - Healthcare Telemedicine Platform

Project Overview: Secure telemedicine platform used for institutional EKG review and healthcare workflows, with integrated payments and protected medical data handling.

Key Contributions:

- Developed secure containerized backend and application infrastructure.
- Implemented data storage and application workflows using Supabase and TrueVault.
- Designed secure workflows for medical data processing and institutional EKG reviews.
- Integrated Stripe APIs for payment processing and reconciliation.
- Automated payment reconciliation for 10,000+ monthly medical transaction records.
- Maintained zero reported compliance breaches during the project period.

Technologies: Python, FastAPI, Docker, Supabase, TrueVault, Stripe API

TATA CONSULTANCY SERVICES

Software Analyst | November 2020 - November 2023

Project: Incentive Management System - FIAT Automobile

Project Overview: Enterprise incentive and commission management platform supporting dealer incentive calculations, sales configurations, transaction processing, and financial reconciliation.

Key Contributions:

- Developed and maintained enterprise microservices using Java and Spring Boot.
- Designed business workflows for incentive and commission processing.
- Implemented dynamic sales configuration functionality.
- Built and maintained CI/CD pipelines using Jenkins and AWS.
- Supported processing and reconciliation of more than \$50M in annual dealer incentives and commissions.
- Worked on production support, defect resolution, performance optimization, and system maintenance.

Technologies: Java, Spring Boot, Microservices, AWS, Jenkins, PostgreSQL

Project: Optical Fiber Management System - Comcast

Project Overview: Telecom network management platform used to monitor and manage optical fiber infrastructure and network traffic across high-throughput telecom nodes.

Key Contributions:

- Developed backend functionality using Express.js and JavaScript.
- Worked on Angular-based enterprise application components.
- Optimized complex MongoDB aggregation pipelines.
- Improved query performance for high-volume telecom data.
- Reduced database query latency by approximately 60%.
- Supported high-availability operations for critical telecom infrastructure maintenance.

Technologies: Node.js, Express.js, JavaScript, Angular, MongoDB

EDUCATION

Academy of Technology (WBUT)

Bachelor of Technology (B.Tech) in Electrical Engineering

2016 - 2020 | CGPA: 8.1